

LARNE, United Kingdom

WMO: 039280

Lat: 54.852N

Lon: 5.831W

Elev: 3

StdP: 101.29

Time Zone: 0.00 (EUW)

Period: 86-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.4	1.4	-3.1	2.9	2.3	-2.0	3.2	3.0	20.6	5.4	17.7	6.1	3.9	240	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	20.6	16.8	19.4	16.0	18.2	15.3	17.6	19.8	16.6	18.7	15.7	17.6	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	11.8	19.0	15.5	11.0	17.9	14.7	10.5	17.0	49.4	20.1	46.5	18.7	43.9	17.8	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.5	12.6	DB	-0.9	23.6	1.1	1.8	-1.7	25.0	-2.3	26.0	-3.0	27.0	-3.7	28.4	(4)
(5)			WB	-2.0	19.5	1.6	1.8	-3.1	20.8	-4.1	21.8	-5.0	22.8	-6.2	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	5.9	5.7	7.1	8.0	10.9	13.0	14.8	14.8	13.0	10.9	8.5	6.6	(6)	
	DBStd	3.90	2.26	2.33	2.26	2.09	2.14	1.93	1.75	1.88	1.80	2.46	2.18	2.63	(7)	
	HDD10.0	607	128	121	93	66	15	1	0	0	1	20	54	108	(8)	
	HDD18.3	3057	386	353	347	309	232	161	111	111	159	231	296	362	(9)	
	CDD10.0	595	1	1	4	7	41	91	149	149	91	47	9	4	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	4	0	0	0	0	0	0	3	1	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.5	6.9	6.7	6.1	5.6	4.9	4.2	4.0	4.3	4.9	5.8	6.2	6.3	(14)	
(15) Precipitation	PrecAvg	1117	117	90	79	76	76	78	80	85	83	122	114	118	(15)	
	PrecMax	1435	200	188	175	134	215	185	182	230	193	225	226	241	(16)	
	PrecMin	890	31	22	14	17	9	28	25	11	0	38	58	59	(17)	
	PrecStd	134	41	42	36	30	43	43	34	48	47	53	43	42	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	12.7	14.2	16.2	18.8	21.3	23.0	22.3	19.5	17.6	14.6	13.8	(19)	
		MCWB	11.2	10.4	11.2	13.3	14.7	16.1	18.1	18.6	15.8	14.9	13.3	12.5	(20)	
	2%	DB	11.5	11.6	12.7	14.1	16.8	18.9	20.7	20.5	18.0	16.1	13.5	12.7	(21)	
		MCWB	10.0	9.7	10.2	11.9	13.4	15.0	16.8	17.1	15.2	14.2	12.0	11.4	(22)	
	5%	DB	10.5	10.5	11.6	12.5	15.4	17.4	19.4	19.2	16.8	15.2	12.6	11.6	(23)	
		MCWB	9.1	8.8	9.6	10.3	12.4	14.2	15.9	16.3	14.4	13.5	11.2	10.3	(24)	
	10%	DB	9.7	9.5	10.7	11.4	14.2	16.2	18.0	18.1	16.0	14.4	11.8	10.6	(25)	
		MCWB	8.4	7.7	8.8	9.4	11.7	13.7	15.3	15.6	14.0	12.8	10.5	9.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	10.7	11.9	14.0	15.2	16.6	19.2	19.4	16.6	15.4	13.7	13.2	(27)	
		MCDB	12.6	12.0	13.7	15.5	17.8	19.6	22.1	21.8	18.3	16.7	14.4	13.7	(28)	
	2%	WB	10.2	10.0	10.6	12.0	14.0	15.5	17.5	17.9	15.6	14.6	12.4	11.5	(29)	
		MCDB	11.2	11.5	12.2	14.2	16.2	18.2	20.0	19.9	17.3	15.9	13.3	12.5	(30)	
	5%	WB	9.4	9.0	9.8	10.6	12.8	14.7	16.4	16.9	15.0	13.9	11.5	10.5	(31)	
		MCDB	10.4	10.6	11.3	12.1	14.9	16.9	18.7	18.8	16.5	15.2	12.4	11.4	(32)	
	10%	WB	8.6	7.9	9.1	9.7	12.0	13.9	15.5	15.9	14.2	13.0	10.6	9.5	(33)	
		MCDB	9.6	9.3	10.5	11.1	14.0	16.0	17.5	17.6	15.7	14.2	11.6	10.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	4.0	4.3	4.5	5.0	4.7	4.7	5.0	4.7	4.0	3.6	3.5	(35)	
		MCDBR	4.7	5.1	5.2	6.0	7.0	6.6	7.1	6.5	5.5	4.7	4.4	4.0	(36)	
	5% WB	MCWBR	4.6	4.7	4.1	4.3	4.7	4.1	4.4	4.2	4.1	4.0	4.2	3.7	(37)	
		MCWBR	4.7	5.0	5.0	5.7	6.6	6.3	6.5	6.2	5.4	4.6	4.3	4.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.337	0.348	0.372	0.387	0.390	0.384	0.386	0.370	0.364	0.348	0.329	0.326	(40)	
		taud	2.239	2.320	2.299	2.298	2.338	2.390	2.390	2.448	2.455	2.427	2.308	2.204	(41)	
	Ebn at Noon		574	724	793	836	854	863	853	847	801	721	585	492	(42)	
		Edh at Noon	61	79	100	114	116	111	109	97	85	70	56	51	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.54	1.23	2.36	3.83	4.89	4.92	4.65	3.81	2.66	1.47	0.70	0.40	(44)	
	RadStd		0.06	0.12	0.19	0.30	0.43	0.49	0.40	0.29	0.17	0.16	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page